



RIPHAH
INTERNATIONAL UNIVERSITY

BS-AI

LAB TASK 09

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Sap I'D: 72405

Subject: Object Orientated
Programming

Teacher: Sir Imran Khan

Task 1:

```
package com.mycompany.lab9;

public class Lab9 {
    public static void main(String[] args) {
        System.out.println("Name: Abdullah");
        System.out.println("SAP ID: 72405\n");

        Shape s1 = new Circle(5);
        s1.draw();
        System.out.println("Circle Area: " + s1.calculateArea());

        Shape s2 = new Cylinder(3, 7);
        s2.draw();
        System.out.println("Cylinder Surface Area: " + s2.calculateArea());
    }
}

class Shape {
    void draw() {
        System.out.println("Drawing a generic shape...");
    }

    double calculateArea() {
        System.out.println("Area calculation not defined for generic shape.");
        return 0;
    }
}

class Circle extends Shape {
```

```

class Circle extends Shape {
    double radius;

    Circle(double radius) {
        this.radius = radius;
    }

    @Override
    void draw() {
        System.out.println("Drawing a Circle...");
    }

    @Override
    double calculateArea() {
        return Math.PI * radius * radius;
    }
}

```

```

class Cylinder extends Shape {
    double radius;
    double height;

    Cylinder(double radius, double height) {
        this.radius = radius;
        this.height = height;
    }

    @Override

```

```
double radius;  
double height;
```

```
Cylinder(double radius, double height) {  
    this.radius = radius;  
    this.height = height;  
}
```

```
@Override  
void draw() {  
    System.out.println("Drawing a Cylinder...");  
}
```

```
@Override  
double calculateArea() {  
    // Total surface area =  $2\pi r(h + r)$   
    return 2 * Math.PI * radius * (height + radius);  
}
```

```
}
```